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Elasticity, deformation and fracture of mixed fluoride–phosphate glasses

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ABSTRACT

Fluoride–phosphate (FP) glasses have been developed (in terms of glass forming ability) as a more stable alternative to fluoride glasses. Benefitting from the low polarizability of the fluoride anion, high electronic band-gap and low phonon energy, they present one of the most important host species for the inclusion of rare-earth and other optically active dopants into fiber and bulk glass. A major limitation, however, has been the mechanical performance of these glasses and their susceptibility to damage and fracture. Here, we provide comprehensive data on elasticity, plasticity and fracture of a series of mixed fluoride–phosphate glasses with strongly ionic bonding character, spanning the compositional join of strontium metaphosphate and alkaline earth aluminum fluoride. Simplistic models such as that of Makishima and Mackenzie can reproduce the mechanical properties of these glasses within a limited compositional range. This break-down is explained in the context of glass structure, in particular, the role of intermediate $^{VI}Al^{3+}$ species which act as bridges between the fluoride and the phosphate sub-networks. A simple binominal model is implemented to estimate the atomic fractions of aluminum as well as of magnesium cations, which are linked to two or more phosphate units, and used as an indicator for the connectivity between the anion sub-networks. Such adjustment of the Makishima–Mackenzie model provides an unambiguous rational for the elastic properties, but also for the susceptibility to inelastic deformation and fracture of FP glasses.

1. Introduction

The mechanical properties remain one of the major limiting factors in almost any application of brittle glasses [1]. This is primarily due to the high defect susceptibility, which easily leads to creation of Griffith flaws and, in consequence, fracture. High strength is therefore often associated with high resistance against the initiation and growth of microscopic flaws and cracks upon sharp (or blunt) contact [1–3]. There is strong experimental evidence for a direct link between crack initiation and the degree of inelastic deformation in oxide glasses, where both mechanisms can be tailored within certain limits by adjusting the chemical composition [4–9]. Such modifications are commonly achieved by varying the cationic species within the glass network, but a partial substitution of O^{2-} by N^{3-} or C^{4-} can affect the mechanical properties of oxide glasses in a broad way, too, e.g., by introducing stronger bonds, by increasing the volume density of bonding energies or by changing the network dimensionality [10,11]. These effects are now well-described for many covalent glasses. However, only little is known about the effects which control the mechanical properties of ionic fluoride and fluoride–phosphate glasses which in bulk or fiber form represent a major class of materials for high-power glass lasers.

Fluoride glasses were initially developed for their optical properties, making use of the low atomic polarizability of the fluoride anion, which leads to low refractive index and high anomalous partial dispersion. Due to the high resonance frequency of the fluoride bonds, low polarizability, and the, potentially, high mass of cations, this group of glasses further enables high optical transparency in the UV as well as in the infrared spectral region [12–17]. Finally, fluoride glasses typically facilitate high rare earth solubility and low phonon energy. These features rank fluoride glasses among the most promising materials for potential applications in photonic devices, such as laser host materials or components of IR optics [14–16,18,19]. In contrast, their practical relevance is compromised on the one hand by difficulties concerning the preparation of these glasses, due to the low viscosities of fluoride melts, their high crystallization tendency during cooling, and their reactivity with atmospheric air [12,17,20,21]. On the other hand, the poor mechanical stability of fluoride glasses, poses another major limitation to certain applications [10,19]. The former problem has been treated with the introduction of minor amounts of phosphates, which positively affect the glass-forming ability in mixed fluoride–phosphate glasses in comparison to the pure fluoride glasses, without notably compromising the desired optical properties [12,13,17,21]. Regarding the latter issue, studies on the mechanical properties (elastic modulus, fracture toughness or thermal shock resistance) of mixed fluoride–phosphate glasses are scarce, despite their importance in the implementation of these materials within photonic applications [18,22,23]. For example, Djouama et al. [24] investigated the structure and properties of glasses in the system $(80 - x)NaPO_3$–$xMnF_2$–$20ZnF_2$ and reported an increase in the elastic modulus and hardness as $NaPO_3$ is gradually replaced by MnF_2.

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An increasing hardness with increasing fluoride concentration has also been observed in (100 − x)$LiPO_3$–xLiF [25] and (50 − x)CaO–xCaF_2–50P_2O_5 glasses [26]. At the same time, a marked decrease in the indentation fracture toughness and increasing brittleness was noticed for the latter system [26]. Wang et al. [19] studied the effect of small amounts of $Al(PO_3)_3$ on the physical, chemical, and optical properties of a multicomponent fluoride glass, reporting a slight increase in the hardness with increasing phosphate content.

Here, we provide a comprehensive study of the mechanical properties of mixed fluoride–phosphate glasses in the system $Sr(PO_3)_2$–MgF_2–CaF_2–SrF_2–AlF_3, using ultrasonic echography and instrumented nano- and microindentation. These glasses were developed for their optical properties and are well studied in this regard [12,13,17]. Using previously obtained structural data [12,13,20,21,27,28], we further provide a topo-chemical rational for the observed properties, in which fluoride–phosphate glasses can, on the one hand, be used as model systems with predominantly ionic bonding. On the other hand this knowledge can be used for tailoring the mechanical properties through chemical adjustments.

2. Experimental

2.1. Glass preparation

Mixed fluoride–phosphate glasses (denoted FPx, where x refers to the $Sr(PO_3)_2$ concentration in mol%), with compositions ranging from the pure strontium metaphosphate to an alkaline earth fluoroaluminate glass were prepared from high purity, optical grade materials ($Sr(PO_3)_2$, MgF_2, CaF_2, SrF_2 and AlF_3) according to Refs. [13,17]. Their nominal compositions are listed in Table 1 and are schematically depicted in Fig. 1. Noteworthy, within 45 < x < 65, glass formation is largely prevented by the high crystallization tendency of these melts [12,13,21], so that this compositional region has been excluded from the present study. All glasses were melted in batches of 100 g for 90 min at temperatures between 800 and 1200 °C in an electrically heated furnace using a Pt crucible covered with a Pt lid, except for the compositions with $Sr(PO_3)_2$ concentrations ranging from 60 to 100 mol%, which were melted in covered silica crucibles. Glass melts with $Sr(PO_3)_2$ contents up to 3 mol% were quenched between two copper plates, while the remaining glass melts were cast into graphite molds which were preheated at 50 K above the respective glass transition temperature, T_g. The as-melted glasses were subsequently transferred to a cooling furnace and annealed for 1 h at $T_g + 50$ K before cooling down to room temperature with a rate of 30 K/h [21,27]. Values of T_g were determined by differential scanning calorimetry (DSC) and the density ρ was measured with the Archimedes method in distilled H_2O or pure ethanol. All experiments were performed in air under ambient conditions ($T = 24 \pm 2$ °C, $rH = 37 \pm 3$%). All glasses have been studied for fluorine loss, which is negligible for low phosphate glasses, but becomes dominant for high $Sr(PO_3)_2$ levels [12]. Oxygen substitution accompanies the fluoride loss (as HF or POF_3) [12,20].

Table 1
Nominal composition (mol%) of the mixed fluoride–phosphate (FP) glasses investigated, according to Refs. [20,21].

Sample	$Sr(PO_3)_2$	MgF_2	CaF_2	SrF_2	AlF_3
FP00	0	10.0	28.3	23.1	38.6
FP01	1.0	9.5	28.2	22.9	38.4
FP02	2.0	9.5	27.8	22.9	37.8
FP03	3.0	9.5	27.5	22.5	37.5
FP04	4.0	9.5	27.5	21.5	37.5
FP06	6.0	9.5	27.0	21.0	36.5
FP08	8.0	10.0	30.2	16.8	35.0
FP10	10.0	10.0	30.0	15.0	35.0
FP15	15.0	10.0	23.4	19.4	32.2
FP20	20.0	10.0	21.8	18.2	30.0
FP30	30.0	10.0	19.0	15.0	26.0
FP40	40.0	10.0	15.5	13.0	21.5
FP70	70.0	4.0	8.0	7.0	11.0
FP80	80.0	2.5	5.5	4.5	7.5
FP90	90.0	1.5	2.9	1.9	4.0
FP100	100.0	0	0	0	0

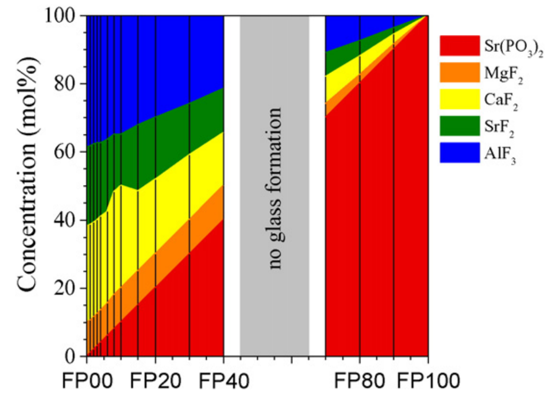

Fig. 1. Nominal compositions (marked with black lines) of the mixed fluoride–phosphate (FP) glasses investigated. The gray bar marks the narrow compositional region, where no glass-formation can be achieved by conventional melt quenching [12,13,17,21].

2.2. Ultrasonic echography

The elastic properties were analyzed using an ultrasonic echometer (model 1077, Karl Deutsch GmbH & Co KG). Here, the transmission times of longitudinal t_L and transversal t_T sound waves were determined with an accuracy of ± 1 ns, using a piezoelectric transducer (operating at f = 8–12 MHz). All measurements were conducted on co-planar, optically polished glass specimen. Values of the longitudinal wave velocity $c_L = 2d/t_L$, and of the transversal wave velocity $c_T = 2d/t_T$ were derived from the sample thickness d, which was determined with an accuracy of ± 2 μm using a micrometer screw, and the time separating two consecutive echoes. The elastic constants, including the shear modulus G, bulk modulus K, elastic modulus E, as well as the Poisson ratio ν, were calculated from the following relationships [29]:

$$G = \rho \cdot c_T^2 \tag{1}$$

$$K = \rho\left(c_L^2 - \frac{4}{3}c_T^2\right) \tag{2}$$

$$E = \rho\left(\frac{3c_L^2 - 4c_T^2}{(c_L/c_T)^2 - 1}\right) \tag{3}$$

$$\upsilon = \frac{c_L^2 - 2c_T^2}{2(2c_L^2 - c_T^2)}. \tag{4}$$

2.3. Nanoindentation

The hardness H and elastic modulus E were characterized through instrumented indentation testing using a nanoindenter platform (G200, Agilent Inc.), equipped with a three-sided Berkovich diamond tip and operating in the continuous stiffness measurement (CSM) mode. With this setup, the values of H and E were continuously recorded as a function of the displacement into the surface by applying a weak oscillation ($\Delta h = 2$ nm, $f = 45$ Hz) to the indenter tip [30]. On every glass specimen, 15 indents with a depth of 2 μm were generated at a constant strain-rate $\dot{\varepsilon}$ of 0.05 s^{-1}.

The strain-rate sensitivity m was determined in a nanoindentation strain-rate jump test according to Maier et al. [31], previously implemented for oxide glasses [30]. In this experiment, a Berkovich diamond tip penetrates the glass surface, initially at a constant strain-rate of 0.05 s^{-1} until the displacement into the surface exceeds a value of 500 nm. Subsequently, reversible strain-rate jumps are applied in intervals of 250 nm, while changes of the hardness are determined with the CSM equipment ($\Delta h = 5$ nm, $f = 45$ Hz) [30]. On each sample, 10 strain-rate jump tests were conducted with strain-rates of 0.05; 0.007 and 0.001 s^{-1}. The strain-rate sensitivity was derived from the slope of the linear regression between the logarithm of the hardness, lnH, and the logarithm of the indentation strain-rate, $ln\dot{\varepsilon}_i$ [32,33]:

$$m = \frac{\partial lnH}{\partial \ln \dot{\varepsilon}_i} \tag{5}$$

where $\dot{\varepsilon}_i$ is equal to $\dot{\varepsilon}/2$ for materials with a depth-independent hardness [34].

2.4. Microindentation

The Vickers hardness H_V was determined with a microhardness tester (Duramin-1, Struers GmbH). On every glass specimen, 15 indents with a peak load P of 981 mN (=100 g) were created and the corresponding values of H_V were calculated from the length of the projected indentation diagonals d by means of the following equation:

$$H_V = 1.8544 \frac{P}{d^2}. \tag{6}$$

The indentation fracture toughness K_c was estimated via Vickers indentation, according to Anstis et al. [35]:

$$K_c = 0.016 \left(\frac{E}{H_V}\right)^{1/2} \frac{P}{c^{3/2}} \tag{7}$$

where c is the distance between the tip of a median-radial crack and the center of the residual Vickers hardness imprint. To ensure the formation of half-penny shaped median–radial cracks ($c/a > 2.5$, where a is equal to one half of the indentation diagonals), loads of 4.91 N (=500 g) and 9.81 N (=1 kg), in the case of $Sr(PO_3)_2$, respectively, were applied with the microhardness tester and the values of K_c were averaged over at least 25 indents. To be noted at this point, the thus obtained value of K_c does not represent fracture toughness, but is taken only as an empirical indicator for comparing the crack resistance within the present group of samples. Clearly, K_c strongly depends on testing conditions, e.g., testing environment, and cannot, per se, be taken as a general property for comparison over multiple material classes and testing situations.

The empirical brittleness parameter B reflects the relationship between the resistance against plastic deformation, H_V, and fracture K_c [36].

$$B = \frac{H_V}{K_c} \tag{8}$$

3. Results

The experimental data on the FP glasses investigated in the current study are summarized in Table 2 and the compositional dependencies of selected mechanical properties are presented in Fig. 2 as a function of the $Sr(PO_3)_2$ content. The value of E, as determined through ultrasonic echography, initially increases with increasing fluoride content and exhibits a maximum ($E = 78.5 \pm 1.3$ GPa) at around 30 mol% $Sr(PO_3)_2$. Afterwards, with the further incorporation of additional fluorides, a progressive decrease of E is observed, whereas the value of the alkaline earth fluoroaluminate glass FP00 ($E = 66.2 \pm 2.0$ GPa) is considerably higher than that of the pure $Sr(PO_3)_2$ glass FP100 ($E = 54.8 \pm 0.3$ GPa). In addition to the ultrasonic echography, as already noted, the values of E were also determined through nanoindentation. Although both experimental techniques display the same compositional trend, a large discrepancy between the absolute values of E is observed. This result is most likely related to the indentation pile-up which occurs at the periphery of the indent during the penetration experiment. Extensive pile-up formation has already been observed in alkali alumino-phosphates [37], and also in a fluorine-based glasses [5]. It is well known that this effect results in an overestimation of E due to an underestimation of the contact area between the indenter tip and the material tested [38]. Finally, a slight but continuous increase occurs in ν when $Sr(PO_3)_2$ is substituted by fluorides. However, ν remains within ~0.3 for glasses with high fluoride concentration.

Regarding plastic deformation and fracture, first, marked differences are detected between the hardness H and the Vickers hardness H_V as determined by nano- and microindentation, respectively. In contrast to E, this result cannot be assigned to pile-up alone. Instead, the large contribution of elastic deformation on the indentation response of glasses also

Table 2
Physical and mechanical properties of the FP glasses investigated. The glass transition T_g was determined by differential scanning calorimetry (DSC) and density ρ was obtained by the Archimedes method in ethanol. The elastic constants, including the shear G, bulk K, elastic modulus E, as well as the Poisson ratio v were characterized by ultrasonic echography. The elastic modulus E, hardness H, and strain-rate sensitivity m were determined through instrumented indentation testing using a nanoindenter. The Vickers hardness H_V, indentation fracture toughness K_c, and brittleness B were investigated with a microhardness tester.

Sample	T_g (°C) (±2 K)	ρ (g/cm^3) (±0.2%)	G (GPa)	K (GPa)	v	E (GPa)[a]	E (GPa)[b]	H (GPa)	m	H_V (GPa)	K_c (MPa $m^{1/2}$)	B ($\mu m^{-1/2}$)
FP00	400[c]	3.42	25.5 ± 0.3	54.5 ± 1.7	0.298 ± 0.016	66.2 ± 2.0	77.5 ± 0.6	4.84 ± 0.05	0.0282	3.75 ± 0.06	0.39 ± 0.03	9.6 ± 0.6
FP01	–	3.54	26.4 ± 0.2	58.7 ± 0.9	0.304 ± 0.007	68.9 ± 1.1	80.9 ± 0.5	5.00 ± 0.04	0.0303	3.83 ± 0.05	0.39 ± 0.03	10.0 ± 0.8
FP02	410[c]	3.45	27.0 ± 0.2	59.3 ± 0.9	0.303 ± 0.007	70.3 ± 1.1	80.9 ± 0.5	5.08 ± 0.02	0.0308	3.89 ± 0.04	0.37 ± 0.02	10.4 ± 0.6
FP03	–	3.46	26.7 ± 0.1	59.5 ± 0.4	0.305 ± 0.002	69.7 ± 0.5	81.7 ± 0.8	5.17 ± 0.03	0.0292	3.95 ± 0.06	0.39 + 0.02	10.3 ± 0.6
FP04	425[c]	3.45	27.0 ± 0.2	60.8 ± 1.1	0.307 ± 0.008	70.6 ± 1.2	82.4 ± 0.4	5.21 ± 0.03	0.0285	4.01 ± 0.07	0.39 ± 0.03	10.5 ± 0.9
FP06	–	3.46	28.0 ± 0.3	62.7 ± 1.8	0.306 ± 0.015	73.1 ± 2.1	82.6 ± 0.3	5.22 ± 0.02	0.0248	4.06 ± 0.06	0.38 + 0.03	10.7 ± 0.7
FP08	–	3.47	27.7 ± 0.2	61.4 ± 1.0	0.304 ± 0.008	72.3 ± 1.2	84.5 ± 0.5	5.41 ± 0.03	0.0242	4.10 ± 0.08	0.39 + 0.02	10.6 ± 0.6
FP10	445[c]	3.40	28.5 ± 0.2	62.9 ± 1.1	0.303 ± 0.008	74.3 ± 1.3	86.1 ± 0.3	5.64 ± 0.04	0.0263	4.27 ± 0.08	0.38 ± 0.02	11.4 ± 0.6
FP15	460[c]	3.48	29.0 ± 0.2	62.9 ± 1.1	0.300 ± 0.008	75.4 ± 1.3	88.0 ± 0.5	5.84 ± 0.05	0.0235	4.35 ± 0.09	0.39 ± 0.02	11.2 ± 0.7
FP20	480[c]	3.51	29.9 ± 0.2	63.8 ± 1.1	0.298 ± 0.008	77.5 ± 1.4	90.3 ± 0.5	6.09 ± 0.04	0.0249	4.58 ± 0.14	0.38 ± 0.02	12.0 ± 0.7
FP30	490[c]	3.51	30.3 ± 0.2	64.5 ± 1.1	0.297 ± 0.008	78.5 ± 1.3	92.5 ± 1.0	6.33 ± 0.05	0.0225	4.77 ± 0.13	0.36 ± 0.03	13.3 ± 0.9
FP40	500[c]	3.46	29.8 ± 0.2	62.3 ± 1.1	0.293 ± 0.008	77.2 ± 1.4	88.9 ± 0.4	6.23 ± 0.04	0.0236	4.70 ± 0.15	0.36 ± 0.02	13.1 ± 0.7
FP70	552	3.26	23.7 ± 0.2	48.9 ± 0.7	0.292 ± 0.006	61.1 ± 0.9	67.1 ± 0.3	5.32 ± 0.04	0.0233	4.17 ± 0.12	0.33 ± 0.03	11.1 ± 0.9
FP80	551	3.25	23.2 ± 0.2	46.9 ± 0.8	0.288 ± 0.007	59.7 ± 1.0	64.7 ± 0.2	5.28 ± 0.03	0.0247	4.07 ± 0.09	0.36 ± 0.03	11.4 ± 1.1
FP90	543	3.20	21.2 ± 0.1	42.6 ± 0.5	0.287 ± 0.004	54.5 ± 0.6	58.5 ± 0.3	4.88 ± 0.03	0.0263	3.79 ± 0.04	0.40 ± 0.03	9.5 ± 0.8
FP100	485[c]	3.18	20.2 ± 0.1	39.5 ± 0.6	0.281 ± 0.006	51.8 ± 0.8	54.8 ± 0.3	4.53 ± 0.03	0.0280	3.61 ± 0.07	0.48 ± 0.04	7.6 ± 0.6

[a] Values of E were determined by ultrasonic echography.
[b] Values of E were determined through nanoindentation.
[c] Values on T_g are taken from Ref. [21].

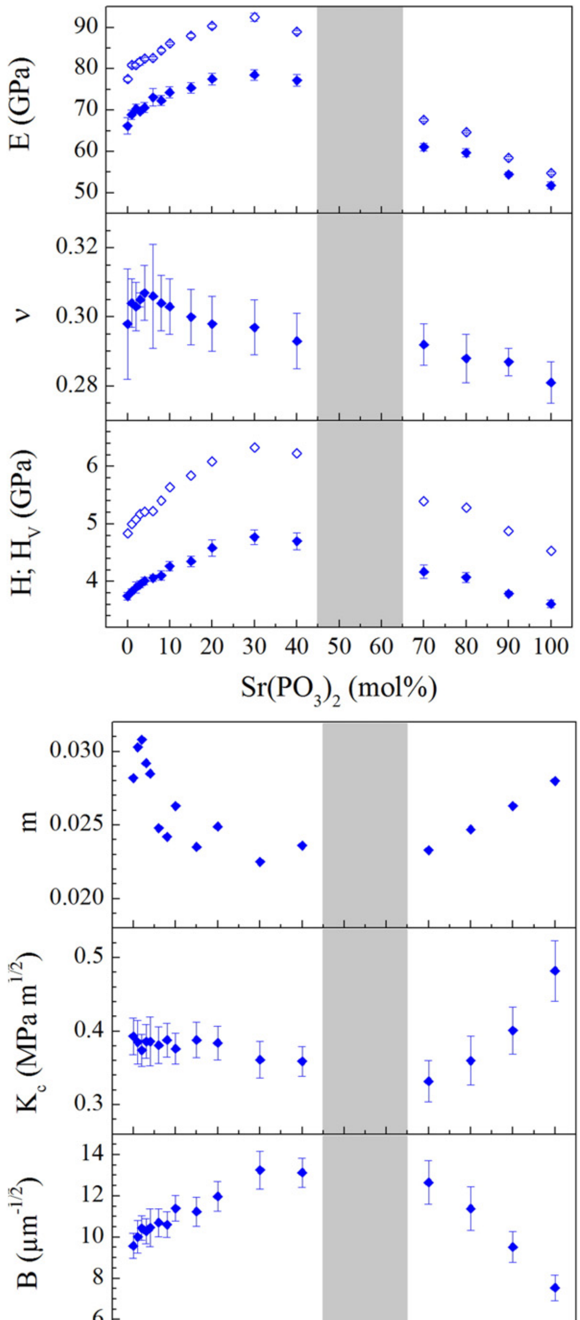

Fig. 2. Compositional dependence of the mechanical properties of the FP glasses on the $Sr(PO_3)_2$ content: elastic modulus E, Poisson ratio ν, hardness H, Vickers hardness H_V, strain-rate sensitivity m, indentation fracture toughness K_c, and brittleness B. Filled symbols in the graphs of E and H display data obtained by ultrasonic echography and Vickers indentation, respectively, while open symbols depict the results determined through nanoindentation. The gray bars mark the narrow compositional region, where no glass-formation can be achieved by conventional melt quenching [12,21].

has to be considered, as it results in substantial deviations between the contact area under load, which determines H, and the size of the residual hardness imprint, which is used to estimate H_V [39]. Nonetheless, both indentation techniques reveal the same compositional trend. With the introduction of fluorides into the $Sr(PO_3)_2$ glass, the resistance against the penetration of an indenter tip increases to reach a maximum ($H = 6.33 \pm 0.05$ GPa; $H_V = 4.77 \pm 0.13$ GPa) at around 30 mol% of $Sr(PO_3)_2$, followed by a continuous decrease until the alkaline earth fluoroaluminate glass is reached, where the values of H and H_V ($H = 4.84 \pm 0.05$ GPa; $H_V = 3.75 \pm 0.06$ GPa) are slightly higher in comparison to the pure metaphosphate glass (FP100: $H = 4.53 \pm 0.03$ GPa; $H_V = 3.61 \pm 0.07$ GPa). The strain-rate sensitivity m initially decreases with increasing fluoride content. Although there is some scatter in the experimental data, a minimum value of m is clearly visible at around 30 mol% $Sr(PO_3)_2$ ($m = 0.0225$). With the incorporation of additional amounts of fluorides, m re-increases, while the maximum value ($m = 0.0308$) was found for the FP02 glass with 2 mol% $Sr(PO_3)_2$.

For the indentation fracture toughness K_c, in addition to the previous comments, it is noted that the presence of multiple cracks as well as the complex residual stress field around the Vickers indents prevent a direct comparison between K_c and the fracture toughness K_{Ic}, which is commonly determined by standardized testing methods [40]. For that reason, K_c is treated solely as an indicator for the compositional dependence of the fracture resistance within this series of FP glasses. Two prominent features are observed in this way. First, the presence of only small amounts of fluorides results in a sudden drop of K_c. Furthermore, a minimum of $K_c = 0.33 \pm 0.03$ MPa $m^{1/2}$ occurs within the phosphate-rich region at around 70 mol% $Sr(PO_3)_2$. Interestingly, this minimum does not coincide with the maxima of E, H and H_V, or the minimum of m, which were all observed at around 30 mol% $Sr(PO_3)_2$.

A first consideration of the brittleness parameter B shows a compositional dependency which is equivalent to the trends found for E, H and H_V, i.e., the introduction of fluorides results in a progressive increase of B up to a maximum value ($B = 13.3 \pm 0.9\ \mu m^{-1/2}$) at around 30 mol% $Sr(PO_3)_2$. At even higher fluoride concentrations, the brittleness of the FP glass decreases down to a value of $B = 9.6 \pm 0.6\ \mu m^{-1/2}$ for the alkaline earth fluoroaluminate glass FP00, which is significantly above the value determined for the pure $Sr(PO_3)_2$ glass FP100 ($B = 7.6 \pm 0.6\ \mu m^{-1/2}$). While the initial increase of B in the phosphate-rich region can be attributed to the increasing resistance against plastic deformation (increasing H_V), on one hand, and the decreasing fracture resistance (decreasing K_c), on the other hand, the decrease of B in the fluoride-rich region is based mainly on the decrease of H_V, since K_c remains relatively constant within this compositional region.

4. Discussion

Depending on the respective scale, the elastic properties of a glass are affected by several different structural aspects. At the atomic scale, the elastic modulus is closely related to the atomic bonding energy and packing density, while at the molecular scale, the network dimensionality and the network topology, e.g., the presence of chains, rings, or layered features, have to be taken into account [10]. Nonetheless, in the past years, various simplistic theoretical models were developed to estimate the elastic modulus of glasses based only on their chemical compositions [8]. Among these approaches, the presumably most frequently applied model was proposed by Makishima and Mackenzie [41], which is based on earlier considerations of ionic crystals. In their approach, the elastic modulus is determined from the atomic packing density C_g and the volume density of bonding energy, which is equal to the dissociation energy per unit volume G_i of every single constituent within the glass, weighted by its molar fraction f_i [41],

$$E = 2C_g \sum f_i G_i \tag{9}$$

where the parameter C_g relates the theoretical molar volume $V_i = 4 / 3\pi N(xr_A^3 + yr_B^3)$ which is occupied by the ions of a compound A_xB_y, with the molar mass M_i, the Avogadro number N and the ionic radii r_A and r_B, respectively, to the effective molar volume of the glass:

$$C_g = \rho \frac{\sum f_i V_i}{\sum f_i M_i}. \tag{10}$$

The values of G_i can be calculated from the molar heat of formation ΔH_f of each constituent and its respective atoms in their gaseous states, according to [42]:

$$G_i = \frac{\rho_i}{M_i}\left[x\Delta H_f(A, gas) + y\Delta H_f(B, gas) - x\Delta H_f(A_xB_y, crystal)\right]. \quad (11)$$

The Makishima–Mackenzie model has been extended to fluoride [22,43] as well as oxyfluoride glasses [44,45] and fairly good agreement was found between the experimentally determined elastic moduli and the predicted values. Therefore, here as well, we use Eq. (9) to calculate the elastic modulus for the present series of FP glasses. Comparing this calculation with experimentally obtained values from the ultrasonic echography enables evaluation of the effects of the volume density of bonding energy and of the atomic packing density on the elastic properties separately from the influences of intermediate-range structural aspects. For G_i, we use: (SrO) = 45.4 kJ/cm^3; (P_2O_5) = 28.2 kJ/cm^3; (MgF_2) = 71.9 kJ/cm^3; (CaF_2) = 63.4 kJ/cm^3; (SrF_2) = 51.7 kJ/cm^3; (AlF_3) = 70.9 kJ/cm^3 [42,45]. Ionic radii r_A are: (Mg^{2+}) = 0.57 Å; (Ca^{2+}) = 1.06 Å; (Sr^{2+}) = 1.21 Å; (Al^{3+}) = 0.54 Å; (P^{5+}) = 0.17 Å, and r_B are: (O^{2-}) = 1.40 Å; (F^-) = 1.33 Å [46]. The obtained results are summarized in Table 3 and shown in Fig. 3 as a function of the $Sr(PO_3)_2$ content.

At first glance, the values of C_g follow the same compositional dependency which was seen before in the experimental data of, e.g., E or H, with only the two glasses FP01 and FP08 falling out of this trend. The exceptionally high value of ρ explains the high C_g value for FP01. For FP08 the explanation is less obvious, changes in the density due to slight deviations in the composition, as this is the transition where the introduction of $Sr(PO_3)_2$ changes from the combined substitution of SrF_2, CaF_2, and AlF_3 to a more pronounced substitution of SrF_2 at higher phosphate concentrations (see Fig. 1). Overall, a more dense packing of the atoms is achieved with the introduction of fluorides into the $Sr(PO_3)_2$ glass matrix. This finding is in accordance with previous observations [19,27,47], where the increase of ρ with increasing fluoride content, and the associated increase of C_g, was attributed primarily to the smaller theoretical molar volume occupied by the fluorine anions F^- (V_i = 5.93 cm^3/mol) compared to the larger phosphate units $[PO_3]^-$ (V_i = 20.78 cm^3/mol). However, the maximum value of C_g at around 30 mol% $Sr(PO_3)_2$ implies that larger fluoride concentration does not necessarily lead to the highest values of C_g. Instead, the incorporation of small amounts of $Sr(PO_3)_2$ seems to promote a high atomic packing within this series of FP glasses, due to the presence of several different ionic species with varying ionic radii. In contrast to the compositional dependence of C_g, it is obvious that the values of Σf_iG_i continuously increase as $Sr(PO_3)_2$ is progressively substituted by the fluorides, since

Table 3
Atomic packing density C_g, volume density of bonding energy Σf_iG_i, and the elastic modulus E, as determined through the model of Makishima and Mackenzie [41], of the FP glasses investigated.

Sample	C_g	Σf_iG_i (kJ/cm^3)	E (GPa)
FP00	0.619	64.44	79.8
FP01	0.641	64.14	82.2
FP02	0.624	63.83	79.7
FP03	0.627	63.59	79.7
FP04	0.628	63.44	79.6
FP06	0.630	62.89	79.3
FP08	0.642	62.78	80.6
FP10	0.634	62.46	79.2
FP15	0.634	60.41	76.6
FP20	0.641	59.05	75.7
FP30	0.646	56.47	73.0
FP40	0.638	53.70	68.5
FP70	0.606	45.13	54.7
FP80	0.606	42.37	51.3
FP90	0.599	39.86	47.8
FP100	0.596	36.80	43.9

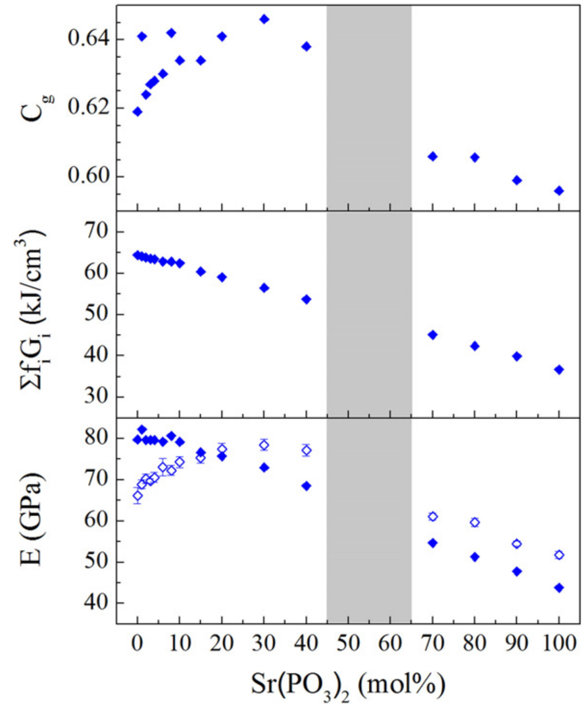

Fig. 3. Compositional dependence of the atomic packing density C_g, volume density of bonding energy Σf_iG_i, and elastic modulus E of FP glasses on the $Sr(PO_3)_2$ content. Filled symbols in the graph of E display the results predicted by the model of Makishima and Mackenzie [41], while open symbols depict the results determined through ultrasonic echography. The gray bars mark the narrow compositional region, where no glass-formation can be achieved by conventional melt quenching [12,21].

the latter exhibit considerably higher values of G_i in comparison to the oxides. It should be noted that older studies on this FP glass series showed that the fluoride loss intensifies from traces (less than 10% relative loss of F for FP10) to significant fractions, as the phosphate content increases (FP40 loses 40%, FP80 even 80% of the fluorides, taking up oxygen from atmospheric reactions instead) [12]. The high volatility of F-species and the difficulties of exact F analyses, are not unique to this studied FP series, but should be considered an important parameter when comparing any structural or property values of fluoride glasses from the literature.

Using the model of Makishima and Mackenzie [41], within the phosphate-rich region, the compositional dependence of E can be reproduced quite well, although the model systematically underestimates the value of E. Within the fluoride-rich region, the errors in the calculations become more pronounced. The aforementioned decrease of E for $Sr(PO_3)_2$ concentrations below 30 mol% can obviously not be described by this simplistic theoretical approach. Instead, the model predicts a continuous increase of E from the strontium metaphosphate glass FP100 to the alkaline earth fluoroaluminate glass FP00, which results in a growing divergence between the measured and the calculated values of E with decreasing $Sr(PO_3)_2$ content. This mismatch appears to be contradictory to the results reported, e.g., by Hager [44] within the system (40 − x)PbO–xPbF$_2$–60B$_2$O$_3$ or by Swarnakar et al. [45] for some fluorine-doped alkaline earth aluminum silico-phosphate glasses. However, in the latter study the values of E were initially strongly overestimated by the model, a discrepancy which was attributed to the formation of additional bonds which were not included in the calculations based on the stoichiometric compositions and therefore leading to significant deviations between the experimentally determined and the predicted values of E [45]. To evaluate this effect in the present series of FP glasses, the structural characteristics at both the atomic and the molecular scale have to be taken into account.

The structure of FP glasses has extensively been studied by NMR, infrared and Raman spectroscopy [12,13,20,21,27]. FP00 forms an ionic three-dimensional network of corner-sharing octahedral $[AlF_6]^{3-}$ groups, whereby glass-formation is facilitated by the presence of a various alkaline earth cations which induce disorder [20,27]. In contrast, the pure $Sr(PO_3)_2$ glass FP100 consists of covalently bound metaphosphate groups $[PO_2Ø_2]^-$ (where O and Ø symbolize bridging and terminal oxygen ions, respectively), which form a polymer-like network of long chains or rings. These chains and rings are linked by ionic bonds between the network modifier ions and the non-bridging oxygen ions of the phosphate tetrahedra [48,49]. With the introduction of fluorides into the $Sr(PO_3)_2$ glass matrix, the phosphate network starts to depolymerize and terminal pyrophosphate $[POØ_3]^{2-}$ units as well as isolated orthophosphate $[PØ_4]^{3-}$ groups are created at the expense of $[PO_2Ø_2]^-$ groups, which are linked to the fluoride sub-network by the formation of $Al(O,F)_6$ octahedral groups [12,20,21]. This gradual shift from a covalent phosphate network towards a more ionic fluoride glass, for example, is mirrored in the continuous decrease of the glass transition T_g with increasing fluoride content [12,20,27]. However, at low fluoride concentration, this effect seems to be initially compensated by the increasing crosslinking ability of the Al^{3+} ions. A stronger crosslinking of the shorter metaphosphate segments by Al^{3+} in comparison to Sr^{2+} strengthens the glass structure and leads to a large increase in T_g up to a maximum value at around 70 to 80 mol% $Sr(PO_3)$ [50]. On this basis, it has been proposed that the connectivity between the phosphate and fluoride sub-networks by $Al(O,F)_6$ groups may affect other glass properties as well, e.g., the density or the resistance against devitrification, whereas the maximum values of ρ were determined for glasses with 20 to 30 mol% $Sr(PO_3)_2$, which are close to the composition with the highest glass-forming ability at around 15 mol% $Sr(PO_3)_2$ [12,20,27].

The interconnectivity of the phosphate and fluoride sub-networks is controlled by aluminum whose first coordination shell comprises a mixture of oxygen and fluorine anions. In order to verify to which extent this topological feature affects the elastic properties we used a simple binominal model to estimate the atomic fractions of $Al(O,F)_6$ groups linked to two or more phosphate units. As an assumption, this calculation implies the absence of any preferential bonding in the present series of FP glasses, which is supported by experimental evidence [20,21,27]. The atomic fraction of an aluminum atom linked to zero $f_{Al\text{-}0P}$, one $f_{Al\text{-}1P}$ and at least two phosphate tetrahedra $f_{Al\text{-}2P}$, respectively, is calculated through the following equations:

$$f_{Al-0P} = \left[\hat{f}_{Mg} + \hat{f}_{Ca} + \hat{f}_{Sr}\right]^6 \tag{12}$$

$$f_{Al-1P} = 6\hat{f}_P \cdot \left[\hat{f}_{Mg} + \hat{f}_{Ca} + \hat{f}_{Sr}\right]^5 \tag{13}$$

$$f_{Al-2P} = 1 - [f_{Al-1P} + f_{Al-0P}] \tag{14}$$

where $\hat{f}_A$ represents the atomic fraction of each cation normalized to all cations present in the glass except aluminum,

$$\hat{f}_A = \frac{f_A}{\sum f_A}. \tag{15}$$

Likewise, this approximation is applied also to the magnesium cation, since the replacement of strontium ($F = 0.29$ Å^{-2}) by network modifier ions with notably higher field strength F (defined as the charge z dived by the sum of the ionic radii squared, $(r_A + r_B)^2$ [51]; for Mg^{2+}, $F = 0.52$ Å^{-2}) presumably affects the physical and mechanical properties of the phosphate sub-network to a considerable extent [52]:

$$f_{Mg-0P} = \left[\hat{f}_{Ca} + \hat{f}_{Sr} + \hat{f}_{Al}\right]^4 \tag{16}$$

$$f_{Mg-1P} = 4\hat{f}_P \cdot \left[\hat{f}_{Ca} + \hat{f}_{Sr} + \hat{f}_{Al}\right]^3 \tag{17}$$

$$f_{Mg-2P} = 1 - \left[f_{Mg-1P} + f_{Mg-0P}\right]. \tag{18}$$

The results of these calculations are presented in Fig. 4. Here, the compositional dependence of the atomic fractions of aluminum f_{Al} and magnesium f_{Mg} is shown, in addition to the relative amounts of aluminum $f_{Al\text{-}2P}$ and magnesium $f_{Mg\text{-}2P}$ species which are linked to at least two phosphate tetrahedra. Both $f_{Al\text{-}2P}$ and $f_{Mg\text{-}2P}$ initially increase with increasing amount of AlF_3 and MgF_2, respectively. Within the fluoride-rich region, $f_{Al\text{-}2P}$ approaches a maximum at around 20 mol% $Sr(PO_3)_2$, followed by a sudden drop with further increasing fluoride content. Assuming that $f_{Al\text{-}2P}$ can be treated as an indicator for the formation of mixed $Al(O,F)_6$ bonds and, hence, the connectivity between the phosphate and fluoride sub-networks, the location of the $f_{Al\text{-}2P}$ peak displays the compositional region of the highest network dimensionality, which coincides very well with previous observations [12,20]. Moreover, as the infrared, Raman, or NMR spectroscopic investigations did not show any evidence for the presence of P–F bonds, the phosphate and the fluoride sub-networks are thought to be linked only by P–O–Al bridging bonds [12,13,20,27,28]. Note that all glasses were prepared under laboratory atmosphere and thus some oxygen from the air will enter the melt, leading to a slight increase of the oxygen content of the fluoride-rich glasses, on the one hand. On the other hand, evaporation of fluoride is only a minor factor in the fluoride-rich glasses, as it does not change the O:F ratio to a large extent. However, in the phosphate-rich glasses the loss of F is indeed significant [12,20]. Taking this into account, a part of the aluminum has to be treated in terms of Al_2O_3 ($G_i = 119.2$ kJ/cm^3) instead of AlF_3 ($G_i = 70.9$ kJ/cm^3 [42]). In consequence, the appearance of a local maximum of $\Sigma f_i G_i$ is supposed to occur within the compositional region of the maximum value of $f_{Al\text{-}2P}$, rather than the continuous increase of $\Sigma f_i G_i$ with increasing fluoride content as shown in Fig. 3. Similar results are expected when MgF_2 ($G_i = 71.9$ kJ/cm^3) is replaced by MgO ($G_i = 90.0$ kJ/cm^3 [42]) although, in contrast to $f_{Al\text{-}2P}$, the peak of $f_{Mg\text{-}2P}$ is slightly shifted towards higher phosphate concentrations and appears at around 40 mol% $Sr(PO_3)_2$. As a result of the adjustment of $\Sigma f_i G_i$, the compositional dependence of E estimated by the model of Makishima and Mackenzie [41] would change, too, whereas the values of E are supposed to deviate from the initially predicted continuous increase presented in Fig. 3 and tend to follow the compositional trends of both C_g and $\Sigma f_i G_i$, similar to the experimentally determined values of E. The maximum which occurs in the fluoride-rich region is then close to the highest values of C_g and $\Sigma f_i G_i$.

With these findings, not only the elastic properties, but also the inelastic deformation and fracture behavior of FP glasses can be explained. During indentation of glasses, the material initially responds with elastic–plastic deformation and the creation of a residual hardness imprint [53]. Permanent deformation can be achieved by either densification

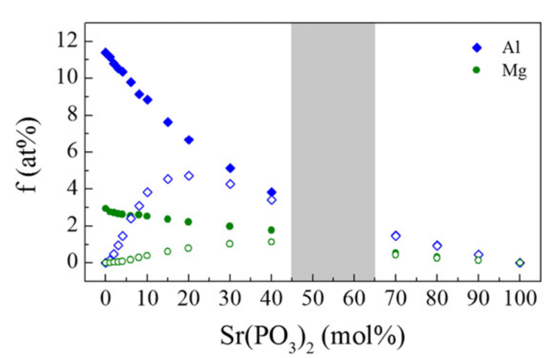

Fig. 4. Compositional dependence of the atomic fractions of aluminum f_{Al} and magnesium f_{Mg} species on the $Sr(PO_3)_2$ content (filled symbols) in the present series of FP glasses. Open symbols represent the fraction of aluminum and magnesium atoms, which are linked to at least two phosphate tetrahedra, $f_{Al\text{-}2P}$ and $f_{Mg\text{-}2P}$, respectively. The gray bars mark the narrow compositional region, where no glass-formation can be achieved by conventional melt quenching [21].

or isochoric shear, depending on the packing of the atoms [54]. Apart from this, a strong correlation between the Poisson ratio and the indentation deformation of glasses was found. Densification occurs preferentially in highly interconnected networks with low values of C_g and ν (such as in vitreous silica), while the deformation of glasses with close-packed structures and large values of ν (such as chalcogenide or metallic glasses) is controlled by shear flow [55,56]. Referring to the current series of FP glasses, the high values of C_g, on the one hand, imply a rather small contribution of densification to the indentation deformation. On the other hand, the large values of ν are a good indicator for inelastic deformation by isochoric shear. Additional support for this assumption is obtained from previous investigations on the indentation response of various alkali alumino-phosphates [37] as well as a fluoride-based glass [5], in which a predominantly shear-mediated indentation deformation was revealed. In this particular case, the phenomenology of hardness can be reduced to the resistance against isochoric shear [56]. Shear flow in glasses is commonly supposed to proceed along the weak ion-bonded interfaces in the network modifier-rich regions [57–59]. Therefore, the initially observed increase of H and H_V with increasing fluoride content is related to the combination of the increase of the volume density of energy (which increases the resistance against shear) and the growing connectivity between the phosphate and fluoride sub-networks (which leads to a reduction of the available sites for shear flow). However, at high fluoride concentrations, this effect is compromised by the progressive depolymerization of the phosphate sub-network, which in fact provides additional paths for a shear-mediated indentation deformation and hence results in the decrease of H and H_V, respectively.

Further support for this interpretation is provided by the results from nanoindentation creep testing. Recently, the strain-rate sensitivity was studied in a variety of glass systems, including covalent, ionic and superionic glasses [8,30]. While high values of m were obtained for glasses with close-packed structures and intermediate network dimensionality, small values of m were found for highly interconnected glass networks with low atomic packing density. Consequently, it was supposed that the strain-rate dependence of the indentation deformation of glasses is the result of localized shear flow [30]. Considering these findings, an alteration of the strain-rate sensitivity is expected in the FP glass series, too, as the resistance against shear-mediated indentation deformation changes with the transition from a pure strontium metaphosphate towards an alkaline earth fluoroaluminate glass. In accordance with the previous results from the hardness measurements, m initially decrease as $Sr(PO_3)_2$ is replaced by an increasing amount of fluorides until the value of m approaches a minimum within the compositional region which corresponds very well to the maximum values of H or H_V. In other words, low values of m correspond to high resistance against localized shear flow. With further increasing fluoride content, shear-mediated indentation deformation is facilitated by the incipient depolymerization of the phosphate sub-network, associated with a re-increase in m.

The prevalent shear-mediated indentation deformation also reflects on the fracture behavior of these glasses. During indentation of a glass, various different crack patterns can occur, e.g., ring, median, radial, or lateral cracks [57]. However, in the evaluation of the defect resistance [60] or indentation fracture toughness [35] only the formation of median and radial cracks is commonly considered, as the propagation of these cracks perpendicular to the glass surface is accompanied by a reduction of the materials strength under tensile or bending loading conditions [61,62]. In principle, the initiation of cracks in an indentation test can be attributed to the presence of tensile stress along the boundary between the plastically deformed volume below the indenter tip and the adjacent elastically strained material. Their formation depends largely on the elastic–plastic properties of the material as well as the mechanism of indentation deformation [63]. For instance, the formation of large radial and lateral cracks can be observed in so-called normal glasses, where the plastic deformation dominates over localized shear flow [57–59]. In contrast, the initiation of ring or cone cracks is favored in anomalous glasses with a large contribution of densification to indentation deformation [57,58,64]. As a result, high values of K_c are usually obtained for anomalous glasses [65,66]. Regarding the present series of FP glasses, their small values of K_c reflect the low resistance against radial cracking. This high sensitivity to mechanically induced defects originates from the preferential shear-mediated indentation deformation, as discussed in the previous paragraphs. Since the brittleness reflects the competition between inelastic deformation and fracture, very brittle glasses are achieved by the gradual replacement of $Sr(PO_3)_2$ with fluorides, and the progressive transition from a covalent phosphate network towards an ionic fluoride glass, similar to previously made observations on, e.g., ionic sodium–zinc–sulfophosphate glasses [67]. However, at intermediate $Sr(PO_3)_2$ content, this effect is superimposed by the aforementioned increase in the resistance against inelastic deformation. The stresses which are induced by the penetration of the indenter tip into the glass surface have to be accommodated subsequently by the formation of radial and lateral cracks, leading to a peak of B at around 30 mol% $Sr(PO_3)_2$.

5. Conclusions

In summary, we provided data on elasticity, plasticity, and fracture of mixed fluoride–phosphate glasses with strongly ionic bonding character. Maxima in Young's modulus ($E = 78.5 \pm 1.3$ GPa), shear modulus ($G = 30.3 \pm 0.2$ GPa), bulk modulus ($K = 64.5 \pm 1.1$ GPa), Berkovich and Vickers hardness ($H = 6.33 \pm 0.05$ GPa; $H_V = 4.77 \pm 0.13$ GPa), as well as in the empirical brittleness parameter ($B = 13.3 \pm 0.9$) are observed at around 30 mol% of phosphate fraction, $Sr(PO_3)_2$, while the Poisson ratio remains close to ν ~0.3 over the whole compositional range, i.e., between the metaphosphate and the fluoride end-members. At the same time, a minimum is observed at this composition in the strain-rate sensitivity ($m = 0.0225$).

Starting from the $Sr(PO_3)_2$ metaphosphate, small amounts of fluoride (and oxide through reactions of F with e.g. atmospheric water) result in a sudden drop of indentation fracture toughness towards a maximum ($K_c =$ which lies, however, still in the phosphate-rich region at around 70 mol% $Sr(PO_3)_2$. In particular, this minimum does not coincide with the above noted extrema.

Simplistic models such as that of Makishima and Mackenzie can reproduce these experimental data only within a limited compositional range. This break-down is explained in the context of glass structure, in particular, the role of intermediate $^{VI}Al^{3+}$ species which act as bridges between the fluoride and the phosphate sub-networks. This involves the formation of $Al(O,F)_6$ mixed-anion octahedral groups. A simple binominal model was employed here to estimate the atomic fractions of aluminum as well as of magnesium cations which are linked to two or more phosphate units. These data were used as indicators for the connectivity between the anion sub-networks. Accordingly, the model of Makishima and Mackenzie is to be adjusted, and an unambiguous rational for the elastic properties, but also for the susceptibility to inelastic deformation and fracture of FP glasses is provided.

The reported findings can help to design optically active glasses and rare-earth host materials with improved mechanical performance such as required for device application, in particular, in optical fiber.

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